

Trends in Mental Disorders in Children and Adolescents Receiving Treatment in the State Mental Health System

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Objective: To examine recent trends in clinical diagnoses of children and adolescents receiving treatment in publicly funded mental health treatment services in the United States.

Method: Data on children and adolescents (≤ 17 years) receiving treatment from publicly funded mental health treatment services recorded in Mental Health Client-Level Data 2013-2021 (total number of records = 13,684,154) were used to examine temporal trends in the proportion of different child and adolescent psychiatric disorders. Trends were examined overall and in age, sex, racial/ethnic, and service strata focusing on community-based programs.

Results: The analyses revealed increases in the proportion of anxiety disorders from 9.6% in 2013 to 19.2% in 2021 (adjusted odds ratio [AOR] 2.17, 95% CI 1.85-2.55, $p < .001$), trauma- and stressor-related disorders from 22.7% to 27.4% (AOR 1.31, 95% CI 1.09-1.57, $p = .004$), and depressive disorders from 13.4% to 17.0% (AOR 1.20, 95% CI 1.03-1.41, $p = .04$). During this same period, the proportion of bipolar disorders declined almost 8-fold from 10.0% to 1.3% (AOR 0.07, 95% CI 0.06-0.09, $p < .001$). The proportion of conduct disorder and oppositional defiant disorder also declined from 9.7% to 4.4% (AOR 0.42, 95% CI 0.32-0.55, $p < .001$) and from 11.1% to 7.8% (AOR 0.79, 95% CI 0.65-0.98, $p = .03$), respectively. Trends varied across sex, age, and racial/ethnic strata.

Conclusion: The composition of childhood psychiatric diagnoses in patients within publicly funded mental health treatment settings changed over the past decade. While some of the trends may reflect changes in diagnostic practices of clinicians, increases in anxiety and depressive disorders parallel trends in the prevalence of these conditions in the general population and highlight a growing need for identifying and treating these conditions in this age group.

Plain language summary: Utilizing the records of services for children and adolescents (≤ 17 years) receiving treatment from publicly funded mental health treatment services that recorded in Mental Health Client-Level Data ($n \sim 13.5$ million), the authors found increases in the proportion of anxiety disorders, trauma- and stressor-related disorders, and depressive disorders between 2013 to 2021. During this same period, the proportion of bipolar disorders, conduct disorder, and oppositional defiant disorder declined. While some of these trends may reflect changes in diagnostic practices of clinicians, increases in anxiety and depressive disorders parallel trends in the prevalence of these conditions in the general population. Results highlight the need for identifying and treating pediatric depression, anxiety, and trauma-related disorders.

Key words: anxiety disorders; bipolar disorders; conduct disorder; depressive disorders; oppositional defiant disorder

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There is growing evidence that the prevalence of mental health problems, including anxiety symptoms, depressive symptoms, and major depressive disorder, have increased among children and adolescents in the United States in the past decade.¹⁻⁹ Concern about these trends is reflected in a recent advisory by the US Surgeon General¹⁰ and a declaration of emergency issued jointly by the American Academy of Pediatrics, American Academy of Child and Adolescent Psychiatry, and Children's Hospital Association.¹¹

While the causes of these trends are widely discussed in the literature,^{1-6,12-15} implications for trends in mental health service use have received less attention. Specifically, few studies have examined trends in

service use among children and adolescents with different psychiatric disorders across a range of public mental health treatment settings. Most past studies have focused on treatment trends in acute care settings (eg, inpatient settings, emergency departments).^{6,16-19} Because children and adolescents who seek treatment in inpatient and emergency department settings typically experience more acute or severe symptoms than children and adolescents in outpatient settings, these studies may not generalize to treatment in outpatient settings.^{1,20-23}

A large population-based survey indicated that there have been recent changes in the clinical composition of young people receiving outpatient mental health care in the

United States. According to the 2005-2018 National Survey on Drug Use and Health report, depression, suicidal ideation, and related internalizing problems accounted for an increasing share of outpatient mental health care of adolescents in the United States, while trouble controlling anger, getting into fights, and other externalizing problems accounted for a declining share of adolescent outpatient mental health care.²⁰ During this period, the percentage of all adolescent patients treated in mental health clinics increased from 12.2% (2005-2006) to 19.3% (2017-2018).¹⁹

Analyses of claims data also report rising rates of diagnosis of young people with anxiety disorders, depressive disorders, and attention-deficit/hyperactivity disorder (ADHD). According to the 2012-2018 MarketScan claims, a large commercial insurance database, the percentage of child and adolescent beneficiaries diagnosed with anxiety disorders increased from 3.7% (2012) to 7.2% (2018), the percentage of children and adolescents diagnosed with depressive disorders increased from 1.8% (2012) to 3.1% (2018), and the percentage of those diagnosed with ADHD increased from 0.9% (2012) to 1.5% (2018).²²

Further examination of treatment seeking for different child and adolescent mental and behavioral disorders in recent years would contribute to better understanding of broad trends in mental health treatment. In the present study, we examined trends in clinical diagnoses of children and adolescents receiving treatment in publicly operated or funded mental health treatment services in the United States between 2013 and 2021, with a focus on community-based programs. These programs are either state operated or receive federal funding through a 2-step block grant process in which the federal government first allocates funds to states, which then have flexibility in determining their own funding priorities.²⁴

Publicly operated and funded mental health programs form an essential component of the mental health care safety net. We characterized trends separately for psychiatric disorders overall and in major sex, age, and racial/ethnic groups. The findings, which are based on a large administrative database, complement past research based on population surveys and commercial claims data. The results further describe recent trends of children and adolescents receiving treatment in state-funded or -operated community-based mental health programs across demographic groups. Findings also have implications for planning services and anticipating the mental health care needs of children and adolescents in the coming years as well as the clinical skills needed by the clinical teams who care for them.

METHOD

Sample

The Mental Health Client-Level Data (MH-CLD) database was used for these analyses. MH-CLD is an administrative database sponsored by the Substance Abuse and Mental Health Services Administration (SAMHSA). MH-CLD captures demographic and mental health characteristics of patients treated in facilities that are either operated by or receive block grant funding through a state mental health agency. Federal block grant legislation requires annual reporting of this information by state mental health agencies to SAMHSA. In this analysis, patients are defined as all enrolled children and adolescents ages ≤ 17 years who received mental health and support services, including screening, assessment, crisis services, or telemedicine during the reporting period. The patient, rather than the treatment episode, is the unit of analysis. MH-CLD collected data on approximately 5 to 6 million patients receiving care in outpatient, hospital inpatient, and residential services each year since 2011. This report is based on approximately 1.5 to 1.9 million children and adolescents each year whose data were collected between 2013 (the first year of MH-CLD publicly available data) and 2021 (the most recent year of data at the time of writing this article) who received treatment in these facilities (total number of records = 13,684,154). Table S1, available online, presents data availability on children and adolescents from each state/jurisdiction in each year. The annual number of participating states ranged from 44 (2016) to 48 (2021). More detailed information regarding MH-CLD is available through SAMHSA publications and previously published research reports using these data.²⁵⁻²⁹ The analyses are based on deidentified publicly available administrative data and exempted from requiring institutional review board approval.

Diagnosis

States were instructed to record the most recent mental health diagnoses that were current during the reporting year. Up to 3 mental health diagnostic codes could be recorded for each client. Diagnostic codes were grouped by MH-CLD investigators into 13 specific diagnostic groups and a group of "other" disorders, based on a modified version of the Clinical Classification Software (CCS).³⁰ Table S2, available online, presents these diagnostic groups and their corresponding *ICD-9/ICD-10* codes.

We limited the CCS diagnostic categories included in this study to 8 categories that were sufficiently prevalent in this age group to allow reliable trend analysis. These included anxiety disorders, depressive disorders, bipolar disorders,

trauma- and stressor-related disorders, ADHD, conduct disorder, oppositional defiant disorder, and pervasive developmental disorders (Table S2, available online). These 8 disorders accounted for 98.1% of child and adolescent patients ages ≤ 17 years. Alcohol and substance use disorders (0.9%); delirium, dementia, and amnestic and other cognitive disorders (0.1%); personality disorders (0.2%); and schizophrenia and other psychotic disorders (0.7%) were not included due to low prevalence in this age group.

Children and adolescents were categorized as having a disorder in these categories if one of the included disorders was recorded as the first, second, or third diagnosis. Because many children and adolescents had diagnoses in more than one category, the diagnostic groups for this study are not mutually exclusive. We additionally conducted sensitivity analysis based on the first recorded diagnosis (see “Analytic Approach”).

The treatment settings in which patients received care were also recorded. Treatment settings included the following:

1. State Mental Health Agency (SMHA)–funded or –operated community-based programs including community mental health centers, outpatient clinics, partial care organizations, partial hospitalization programs, assertive community treatment programs, consumer-run programs, and all community support programs
2. Residential programs including organizations not licensed as a psychiatric hospital whose primary purpose is provision of individually planned mental health treatment services in conjunction with residential care for children and youth
3. State hospitals including all SMHA-funded and -operated organizations operated as hospitals that provide primary inpatient care to individuals with mental illness from a specific geographical area or across a state
4. Other psychiatric inpatient treatment settings including private providers or medical providers licensed and/or contracted through the SMHA including public/private hospitals with separate psychiatric wards
5. Institutions under the justice system including mental health services provided in jails, prisons, and juvenile detention centers

Multiple settings could be recorded if the patient received care from these settings during the reporting period.

Sociodemographic Characteristics

MH-CLD also collected demographic data including age (<12 , 12–14, 15–17 years), sex, and race/ethnicity as

recorded in the administrative records. Race and ethnicity data were combined to create 4 large race/ethnicity groups: Hispanic, non-Hispanic Black, non-Hispanic White, and other racial/ethnic groups. The other racial/ethnic group included American Indian/Alaska Native, Asian, Native Hawaiian or Other Pacific Islander, and some other race alone or more than one race. In further analyses, we also explored variations in trends according to state median income. State median income levels were extracted from Federal Reserve Economic Data files³¹ and divided into tertiles: the low tertile included states with median income of $< \$70,000$, the medium tertile included states with median income $\geq \$70,000$ to $< \$84,000$, and the high tertile included states with median income $\geq \$84,000$.

Analytic Approach

Temporal trends for each diagnostic group were presented as raw percentages in each year. Trends were examined using separate logistic regression models with year as the independent variable and each disorder or disorder group as the dependent variable. Results were presented as unadjusted odds ratios (ORs) and ORs adjusted for sex, race/ethnicity, age group, and service setting. To capture change across the entire 2013–2021 period, consistent with past research,^{1,20} the year variable was converted to range from 0 to 1 by subtracting 2013 from each year and dividing the remainder by 8, thereby testing the trend from 2013 to 2021.

In analyses limited to community-based programs, separate analyses were conducted for each diagnostic group and sex, age, and racial/ethnic strata within each diagnostic group. Interaction terms of time by demographic characteristic were added to evaluate whether diagnostic trends significantly varied across levels of individual demographic characteristics. Regression analyses were conducted with and without adjustment for other demographic variables. In further analyses, trends were examined and compared across state median income level strata within each diagnostic group. Similar interaction tests were conducted to examine whether trends differed across these strata.

The main analyses were followed by a series of sensitivity analyses. In one set of sensitivity analyses, regression models for trends were rerun after excluding states with missing data for most of the years between 2013 and 2021 (ie, Alaska, Georgia, Kansas, Maine, New Hampshire, and West Virginia) or reporting on only a subset of the total individuals served through their SMHAs (New York and Tennessee). In a second set, regression analyses for trends were repeated using only the first recorded diagnosis. A third set of sensitivity analyses examined deviations from linear trend by adding a quadratic term for year to the models (the year variable in these analyses was computed by

subtracting 2013 from the year in these models). Lastly, the potential effect of COVID-19 was examined by introducing linear splines to examine change in level and slope of the trend between 2013-2019 and 2020-2021.

All analyses were conducted using Stata software version 18 (StataCorp LLC, College Station, Texas). Linear splines were created using the `mkspline` command of Stata. The standard errors were adjusted for clustering in states to account for variations among states in diagnostic practices and violation of independence of samples in each state due to the same clients receiving services in multiple years. Statistical significance was set at $p < .05$.

RESULTS

Characteristics of Children and Adolescents

Table 1 presents the demographic characteristics of children and adolescents in the MH-CLD sample between 2013 and 2021. While the majority across all years were male, the proportion of male patients decreased over time (from 58.8% in 2013 to 53.4% in 2021; OR 0.84, 95% CI 0.81-0.87, $p < .001$). Almost half of patients were less than 12 years old. However, the proportion of patients from this younger age group declined over time (from 49.1% in 2013 to 46.2% in 2021; OR 0.92, 95% CI 0.86-0.98, $p = .011$), whereas the proportion of patients in the 12 to 14 years age range increased (from 24.5% in 2013 to 26.6% in 2021; OR 1.11, 95% CI 1.05-1.16, $p < .001$). The proportion of patients in the 15 to 17 years age group did not change appreciably (26.4% in 2013 and 27.2% in 2021). Consistent with demographic trends in the general population,³² an increasing proportion of children and adolescents who received treatment in the public mental health system were of Hispanic ethnicity (from 10.2% in 2013 to 15.4% in 2021; OR 1.55, 95% CI 1.11-2.18, $p = .010$). By contrast, a declining proportion of children and adolescents were from the non-Hispanic White (from 47.8% in 2013 to 39.9% in 2021; OR 0.77, 95% CI 0.62-0.96, $p = .022$) or non-Hispanic Black (from 19.5% in 2013 to 14.4% in 2021; OR 0.71, 95% CI 0.59-0.85, $p < .001$) racial/ethnic groups. The large majority had used community-based programs (98.8%-99.4%). The small proportion that received care at state hospitals or institutions under the justice system declined, although there was considerable variation across the years.

Clinical Diagnoses and Trends in Diagnoses

The most common diagnosis in this sample was ADHD (27.8%), followed by trauma- and stressor-related disorders (24.4%), depressive disorders (14.8%), anxiety disorders

(13.7%), oppositional defiant disorder (10.0%), conduct disorder (7.2%), bipolar disorders (4.9%), and pervasive development disorders (3.8%). In addition, 21.3% had either a diagnosis in one of the categories not included in this study or in “other” diagnoses. These percentages add up to more than 100% because many children and adolescents had more than one recorded diagnosis. Among patients with any of the 8 diagnostic categories, 21.5% had diagnoses in 2 categories, and 4.2% had diagnoses in 3 categories.

Table 2 presents raw percentages, unadjusted ORs, and adjusted ORs (AORs) from trend analyses for the diagnostic categories. Figure 1A, B presents trends based on adjusted percentages from the logistic regression analyses. The proportion of anxiety, depressive, and trauma- and stressor-related disorders increased significantly over the 2013-2021 period. The increase was most marked for anxiety disorders, which increased from 9.6% in 2013 to 19.2% in 2021 (AOR 2.17, 95% CI 1.85-2.55, $p < .001$), followed by trauma- and stressor-related disorders, which increased from 22.7% to 27.4% (AOR 1.31, 95% CI 1.09-1.57, $p = .004$), and depressive disorders, which increased from 13.4% to 17.0% (AOR 1.20, 95% CI 1.03-1.41, $p = .023$).

During this same period, the proportion of bipolar disorders declined almost 8-fold from 10.0% in 2013 to 1.3% in 2021 (AOR 0.07, 95% CI 0.06-0.09, $p < .001$). The proportion of conduct disorder and oppositional defiant disorder also declined from 9.7% to 4.4% (AOR 0.42, 95% CI 0.33-0.55, $p < .001$) and from 11.1% to 7.8% (AOR 0.79, 95% CI 0.65-0.98, $p = .029$), respectively. The trend was also declining for ADHD from 29.8% to 25.3%, but the statistical test for trend across years did not reach a significant level in adjusted analyses. There was no appreciable change in the proportion with pervasive development disorders over the study period.

Trends in Clinical Diagnoses in Different Treatment Settings

Trends were mostly consistent across treatment settings (Table 2). The prevalence of bipolar disorder diagnoses was notably higher in state and other psychiatric hospitals in the earlier years. For instance, nearly 40% of children and adolescents receiving treatment in state psychiatric hospitals and 36.0% of children and adolescents receiving treatment in other psychiatric inpatient settings in 2013 received a diagnosis of bipolar disorder. These proportions were reduced to 6.9% and 5.4%, respectively, in 2021.

Trends in Stratified Analyses

The stratified analyses for trends across strata defined by age groups, sex, and racial/ethnic groups were limited to

TABLE 1 Characteristics of Children and Adolescents Who Received Care in State Mental Health Systems Between 2013 and 2021 and Are Included in Mental Health Client-Level Data (MH-CLD)

Characteristics	Year									Trend across 2013-2021 period		
	2013 (N = 1,636,712)	2014 (N = 1,663,618)	2015 (N = 1,473,157)	2016 (N = 1,610,944)	2017 (N = 1,670,404)	2018 (N = 1,822,515)	2019 (N = 1,891,605)	2020 (N = 1,915,199)	2021 (N = 1,752,884)	OR	95% CI	p
Sex^a	%	%	%	%	%	%	%	%	%			
Female	41.1	41.9	43.0	42.8	43.3	43.4	44.1	44.4	46.5	1.19	1.15-1.23	<.001
Male	58.8	58.0	56.8	56.8	56.4	56.4	55.5	55.3	53.4	0.84	0.81-0.87	<.001
Age, y												
<12	49.1	49.9	48.2	49.0	48.8	49.4	48.6	48.1	46.2	0.92	0.86-0.98	.011
12-14	24.5	24.4	24.8	24.2	24.3	24.5	25.3	25.8	26.6	1.11	1.05-1.16	<.001
15-17	26.4	25.7	27.0	26.7	27.0	26.2	26.1	26.1	27.2	1.01	0.96-1.07	.618
Race/ethnicity^b												
Hispanic	10.2	10.6	11.6	11.2	12.1	12.5	12.1	14.6	15.4	1.55	1.11-2.18	.010
Non-Hispanic Black	19.5	19.7	19.4	18.3	18.5	17.5	17.8	15.3	14.4	0.71	0.59-0.85	<.001
Non-Hispanic White	47.8	47.4	45.0	46.5	46.6	45.6	44.7	42.7	39.9	0.77	0.62-0.96	.022
Other	15.5	15.7	16.3	16.0	16.9	16.7	17.1	17.2	18.0	1.18	0.92-1.50	.186
Type of service^c												
Community-based program	99.4	99.1	98.9	99.1	98.8	99.2	99.2	99.2	99.4	1.10	0.69-1.74	.689
State psychiatric hospital	0.5	0.6	0.6	0.5	0.5	0.5	0.5	0.5	0.4	0.72	0.55-0.94	.016
Other psychiatric inpatient	1.2	1.2	1.6	2.0	2.2	2.0	1.8	1.7	1.5	1.26	0.63-2.52	.515
Residential treatment center	0.8	0.9	0.9	1.0	1.1	1.1	0.9	0.8	0.9	1.02	0.57-1.83	.935
Institution under justice system	0.1	0.5	0.5	0.4	0.4	0.3	0.3	0.2	0.2	0.68	0.55-0.83	<.001

Note: Analyses are based on MH-CLD, 2013-2021. OR = odds ratio.

^aData on sex were missing for 0.1%-0.4% of patients across the years (mean 0.2%).

^bData on race/ethnicity were missing for 6.9%-12.3% of patients across the years (mean 8.2%).

^cTypes of services add up to more than 100% because some children and adolescents received care in more than one service setting.

TABLE 2 Trends in Prevalence of Different Mental Disorders in Children and Adolescents Receiving Treatment in State Mental Health Systems Between 2013 and 2021 According to Treatment Setting

Disorders	Percent of children and adolescents with each disorder receiving care in 2013-2021									Trend across 2013-2021 period					
	2013	2014	2015	2016	2017	2018	2019	2020	2021	Unadjusted			Adjusted ^a		
										OR	95% CI	p	AOR	95% CI	p
All settings															
Anxiety disorders	9.6	10.2	11.8	12.5	14.5	15.9	17.0	16.8	19.2	2.22	1.88-2.62	<.001	2.17	1.85-2.55	<.001
Depressive disorders	13.4	13.8	15.2	14.2	14.3	15.2	16.5	15.7	17.0	1.29	1.11-1.49	.001	1.20	1.03-1.41	.023
Bipolar disorders	10.0	9.6	8.8	5.2	2.6	2.0	1.6	1.4	1.3	0.08	0.06-0.10	<.001	0.07	0.06-0.09	<.001
Trauma- and stressor-related disorders	22.7	22.5	23.3	22.9	24.0	25.6	27.1	25.9	27.4	1.33	1.11-1.58	.002	1.31	1.09-1.57	.004
ADHD	29.8	28.7	27.2	27.5	28.5	29.0	26.7	25.1	25.3	0.83	0.69-0.98	.032	0.90	0.77-1.06	.208
Conduct disorder	9.7	9.4	9.0	7.7	6.7	6.0	5.5	4.9	4.4	0.41	0.31-0.53	<.001	0.42	0.33-0.55	<.001
Oppositional defiant disorder	11.1	10.5	10.2	9.8	10.4	10.3	9.7	8.5	7.8	0.74	0.59-0.93	.010	0.79	0.65-0.98	.029
Pervasive developmental disorders	3.8	4.2	3.2	4.3	4.0	3.9	3.3	3.7	3.8	0.92	0.46-1.88	.827	0.98	0.48-1.99	.949
State mental health agency—funded/operated community-based programs															
Anxiety disorders	9.6	10.2	11.9	12.5	14.5	15.9	17.1	16.9	19.3	2.22	1.87-2.63	<.001	2.18	1.85-2.57	<.001
Depressive disorders	13.3	13.8	15.1	14.1	14.2	15.1	16.4	15.6	16.9	1.28	1.11-1.49	.001	1.21	1.03-1.42	.023
Bipolar disorders	9.9	9.5	8.6	5.1	2.6	2.0	1.6	1.4	1.3	0.08	0.06-0.10	<.001	0.07	0.06-0.10	<.001
Trauma- and stressor-related disorders	22.8	22.6	23.3	22.9	24.0	25.6	27.1	26.0	27.4	1.33	1.11-1.59	.002	1.32	1.10-1.58	.003
ADHD	29.9	28.8	27.3	27.6	28.7	29.1	26.8	25.2	25.4	0.83	0.69-0.98	.032	0.89	0.76-1.04	.152
Conduct disorder	9.6	9.3	8.9	7.7	6.7	6.0	5.5	4.8	4.3	0.40	0.31-0.53	<.001	0.42	0.32-0.55	<.001
Oppositional defiant disorder	11.1	10.5	10.2	9.8	10.4	10.3	9.7	8.5	7.8	0.74	0.59-0.94	.012	0.78	0.64-0.97	.024
Pervasive developmental disorders	3.8	4.2	3.2	4.3	4.1	3.9	3.3	3.7	3.8	0.93	0.46-1.88	.830	0.95	0.46-1.94	.885
State psychiatric hospitals															
Anxiety disorders	10.0	10.9	10.0	15.1	13.7	15.7	17.9	15.7	18.5	2.07	1.27-3.36	.004	2.15	1.37-3.37	.001
Depressive disorders	27.5	28.3	27.9	30.6	28.6	35.1	37.1	31.1	35.5	1.48	0.77-2.83	.237	1.61	0.89-2.94	.117
Bipolar disorders	39.7	33.9	29.0	22.8	10.8	10.8	8.9	7.1	6.9	0.08	0.05-0.11	<.001	0.08	0.05-0.11	<.001
Trauma- and stressor-related disorders	18.4	19.4	17.7	20.9	19.7	22.4	23.1	23.6	21.6	1.38	0.91-2.10	.130	1.38	0.93-2.04	.110
ADHD	25.6	23.6	20.8	23.4	23.3	24.6	26.7	32.0	21.4	1.23	0.72-2.08	.451	1.22	0.79-1.89	.374
Conduct disorder	15.5	13.1	10.8	11.3	10.0	11.0	10.5	8.9	7.3	0.54	0.30-0.97	.039	0.55	0.32-0.94	.029
Oppositional defiant disorder	16.0	14.5	12.7	13.8	12.2	12.6	13.0	13.5	8.0	0.69	0.42-1.14	.151	0.65	0.43-0.99	.045
Pervasive developmental disorders	3.7	3.2	2.5	3.0	2.7	3.0	3.0	3.9	2.4	0.94	0.58-1.52	.801	0.91	0.57-1.46	.689

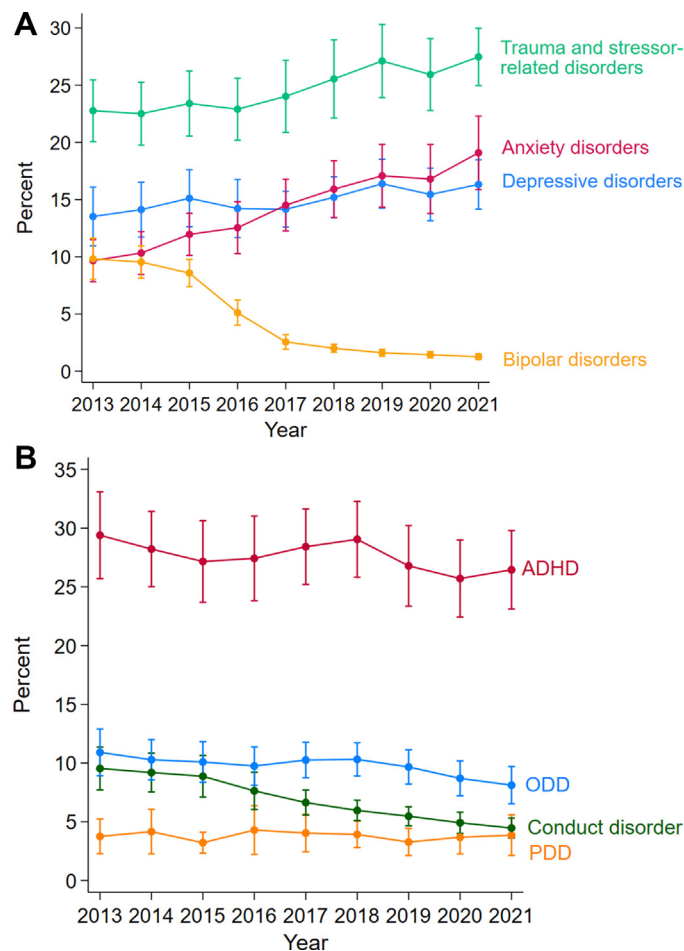
(continued)

TABLE 2 Continued

Disorders	Percent of children and adolescents with each disorder receiving care in 2013-2021									Trend across 2013-2021 period					
	2013	2014	2015	2016	2017	2018	2019	2020	2021	Unadjusted			Adjusted ^a		
										OR	95% CI	p	AOR	95% CI	p
Other psychiatric inpatient settings															
Anxiety disorders	11.1	14.0	15.8	15.7	17.3	21.3	22.5	20.4	23.9	2.24	1.44-3.51	<.001	2.14	1.38-3.33	.001
Depressive disorders	33.8	36.9	38.5	36.4	33.7	39.1	40.9	39.7	43.7	1.39	1.04-1.85	.026	1.37	1.05-1.77	.019
Bipolar disorders	36.0	30.5	25.9	17.3	10.2	8.0	5.8	5.0	5.4	0.06	0.04-0.08	<.001	0.06	0.04-0.08	<.001
Trauma- and stressor-related disorders	18.6	20.5	22.9	22.6	22.8	22.8	24.6	21.1	24.8	1.23	0.83-1.82	.302	1.19	0.80-1.75	.391
ADHD	22.7	24.0	21.8	24.7	24.1	25.1	21.8	20.4	19.8	0.83	0.62-1.11	.201	0.83	0.62-1.11	.213
Conduct disorder	7.3	7.8	6.8	6.4	6.7	5.8	5.1	4.9	3.9	0.53	0.43-0.66	<.001	0.55	0.43-0.68	<.001
Oppositional defiant disorder	12.0	11.5	10.4	13.8	12.9	12.9	11.1	8.7	9.2	0.74	0.58-0.96	.022	0.75	0.57-0.99	.042
Pervasive developmental disorders	3.8	4.3	3.8	4.8	3.5	2.6	1.5	1.3	1.6	0.28	0.11-0.74	0.010	0.27	0.10-0.73	0.009
Residential treatment centers															
Anxiety disorders	6.8	7.5	8.7	9.4	11.0	13.8	14.9	12.8	17.4	2.78	2.00-3.87	<.001	2.71	1.98-3.70	<.001
Depressive disorders	17.1	19.0	21.8	21.2	23.2	26.8	25.0	26.4	26.8	1.73	1.22-2.45	.002	1.67	1.18-2.35	.004
Bipolar disorders	30.2	28.8	22.3	18.0	12.3	8.9	6.8	6.2	5.0	0.09	0.05-0.18	<.001	0.09	0.05-0.17	<.001
Trauma- and stressor-related disorders	23.2	21.8	24.0	23.0	30.3	30.2	28.3	24.5	32.4	1.54	1.03-2.32	.037	1.53	1.03-2.28	.037
ADHD	24.2	26.8	22.4	27.9	27.4	30.4	28.9	22.9	27.4	1.12	0.76-1.66	.569	1.17	0.77-1.79	.464
Conduct disorder	16.3	14.0	12.9	13.0	12.7	12.4	13.3	9.0	9.6	0.61	0.37-1.02	.060	0.63	0.40-1.00	.051
Oppositional defiant disorder	15.3	15.2	13.5	18.7	17.6	18.0	19.1	14.2	14.6	1.04	0.67-1.63	.862	1.05	0.66-1.66	.845
Pervasive developmental disorders	4.4	3.4	3.2	5.1	4.2	2.7	2.7	2.4	2.9	0.59	0.31-1.13	.109	0.61	0.31-1.20	.154
Institutions under the justice system															
Anxiety disorders	5.6	4.1	5.2	6.0	9.2	10.8	11.7	9.6	9.3	3.00	2.21-4.07	<.001	3.14	2.25-4.38	<.001
Depressive disorders	12.3	16.5	17.6	17.1	14.2	16.1	18.0	16.6	15.9	1.05	0.91-1.21	.511	1.04	0.91-1.20	.560
Bipolar disorders	10.0	13.9	16.4	14.8	4.0	3.4	3.0	2.9	1.6	0.09	0.08-0.10	<.001	0.08	0.07-0.10	<.001
Trauma- and stressor-related disorders	16.2	13.6	15.5	16.4	18.4	21.2	25.2	22.6	21.3	2.12	1.53-2.93	<.001	2.17	1.57-3.00	<.001
ADHD	15.7	13.6	13.8	12.9	15.8	16.6	14.4	12.8	12.2	0.96	0.73-1.26	.769	0.97	0.77-1.22	.781
Conduct disorder	14.0	23.3	25.2	23.8	21.4	18.6	21.2	21.4	21.1	0.88	0.70-1.11	.289	0.87	0.70-1.08	.215
Oppositional defiant disorder	12.4	14.1	12.8	11.3	12.0	13.0	12.6	10.6	10.6	0.79	0.68-0.92	.002	0.80	0.69-0.92	.003
Pervasive developmental disorders	0.7	0.3	0.4	0.5	0.8	0.7	0.6	0.6	0.5	1.63	0.99-2.67	.053	1.75	1.18-2.60	.005

Note: Analyses are based on Mental Health Client Level Data, 2013-2021. ADHD = attention-deficit/hyperactivity disorder; AOR = adjusted odds ratio; OR = odds ratio.

^aTrends were examined using logistic regression models adjusting for sex, race/ethnicity, and age group. The model for all settings was additionally adjusted for specific setting type.

FIGURE 1 Prevalence Trends of Child and Adolescent Psychiatric Disorders

Note: Trends in prevalence ($\pm$ 95% CI) of anxiety disorders, depressive disorders, bipolar disorders, and trauma- and stressor-related disorders (A) and in attention-deficit/hyperactivity disorder (ADHD), conduct disorder, oppositional defiant disorder (ODD) and pervasive developmental disorders (PDD) (B) in children and adolescents who received care at state mental health systems between 2013 and 2021. Analyses are based on the Mental Health Client-Level Data, 2013-2021. Please note color figures are available online.

SMHA-funded or -operated community-based programs. The time trends varied considerably based on these demographic characteristics. Time trends according to age are presented in Table 3. The increasing trends in depressive disorders were significantly more pronounced in the 12 to 14 and 15 to 17 years age groups compared with the <12 years age group and in anxiety disorders were limited to these older age groups. In turn, the declining trend in bipolar disorders was less pronounced in the 12 to 14 years and 15 to 17 years age groups compared with the <12 years age group, and the declining trend in oppositional defiant disorder was limited to the 12 to 14 and 15 to 17 years age groups.

Time trends according to sex are presented in Table 4. The increasing trends in anxiety and depressive disorders were significantly more pronounced in female than male

children and adolescents, whereas the increase in trauma- and stressor-related disorders and the decrease in bipolar disorders were more pronounced in boys. In contrast, the decreasing trends in ADHD, conduct disorder, and oppositional defiant disorder were limited to girls or significantly more pronounced in girls.

Time trends according to race/ethnicity are presented in Table 5. Hispanic children adolescents experienced a significantly less pronounced increase in anxiety and depressive disorders compared with non-Hispanic White children and adolescents. The increase in depressive disorders was significant only in non-Hispanic White and Black children and adolescents. Similarly, the increase in trauma- and stressor-related disorders was limited to non-Hispanic Black and White children and adolescents, and Hispanic children and adolescents did not experience any increase in

TABLE 3 Trends in Prevalence of Different Mental Disorders in Children and Adolescents Receiving Treatment in Community-Based Programs Between 2013 and 2021 According to Age Group

Percent of children and adolescents with each disorder receiving care in 2013-2021										Trend across 2013-2021 period ^a						
Age groups	2013	2014	2015	2016	2017	2018	2019	2020	2021	Unadjusted			Adjusted			
										OR	95% CI	p	AOR	95% CI	p	Interaction p
Anxiety disorders																
<12 y	9.3	9.5	10.9	10.9	12.2	12.9	13.9	13.3	14.6	1.67	1.36-2.06	<.001	1.66	1.37-2.02	<.001	Reference
12-14 y	9.7	10.5	12.3	13.3	15.6	17.2	18.2	18.3	21.1	2.43	2.07-2.87	<.001	2.38	2.03-2.80	<.001	<.001
15-17 y	10.1	11.4	13.4	14.9	17.8	20.4	22.0	22.3	25.4	2.98	2.55-3.49	<.001	2.93	2.51-3.41	<.001	<.001
Depressive disorders																
<12 y	5.2	5.0	5.3	4.7	4.6	4.8	5.8	5.1	5.3	1.06	0.88-1.27	.555	1.05	0.87-1.26	.606	Reference
12-14 y	17.5	18.8	20.2	19.0	18.8	20.2	21.8	20.9	22.4	1.29	1.11-1.51	.001	1.20	1.01-1.42	.035	<.001
15-17 y	24.9	26.2	28.2	27.3	27.5	29.9	31.1	30.0	31.6	1.36	1.17-1.57	<.001	1.28	1.09-1.50	.002	<.001
Bipolar disorders																
<12 y	5.9	5.5	5.1	2.8	1.1	0.7	0.5	0.4	0.3	0.04	0.03-0.05	<.001	0.04	0.03-0.06	<.001	Reference
12-14 y	12.1	11.8	10.4	6.1	3.0	2.2	1.7	1.5	1.2	0.06	0.05-0.08	<.001	0.06	0.05-0.08	<.001	<.001
15-17 y	15.3	15.2	13.4	8.7	5.0	4.3	3.6	3.3	2.9	0.11	0.09-0.15	<.001	0.12	0.09-0.15	<.001	<.001
Trauma- and stressor-related disorders																
<12 y	25.8	25.5	26.8	26.2	27.4	29.0	30.9	29.3	31.0	1.33	1.10-1.61	.004	1.32	1.09-1.60	.005	Reference
12-14 y	21.5	21.2	21.7	21.2	22.3	23.8	25.2	24.4	26.0	1.33	1.10-1.60	.003	1.29	1.07-1.55	.008	.654
15-17 y	18.2	17.9	18.5	18.2	19.4	20.8	21.9	21.4	22.5	1.36	1.15-1.62	<.001	1.33	1.12-1.58	.001	.938
ADHD																
<12 y	35.1	33.8	31.9	32.1	33.3	33.5	31.4	29.4	29.1	0.79	0.66-0.96	.018	0.85	0.72-1.00	.055	Reference
12-14 y	30.2	28.6	27.8	28.2	29.3	29.8	26.8	25.5	26.2	0.84	0.70-1.01	.059	0.94	0.80-1.11	.480	.056
15-17 y	19.6	19.1	18.5	18.6	19.5	20.1	18.1	17.2	18.4	0.91	0.75-1.11	.346	1.03	0.86-1.23	.758	.003
Conduct disorder																
<12 y	11.5	11.4	11.2	9.6	8.1	7.2	6.7	5.9	5.5	0.41	0.31-0.54	<.001	0.42	0.32-0.55	<.001	Reference
12-14 y	7.4	7.1	6.7	5.7	5.1	4.7	4.3	3.9	3.4	0.43	0.33-0.57	<.001	0.46	0.35-0.61	<.001	.288
15-17 y	8.3	7.5	6.8	6.0	5.4	4.9	4.3	3.8	3.3	0.39	0.29-0.52	<.001	0.42	0.30-0.56	<.001	.852
Oppositional -defiant disorder																
<12 y	10.0	9.6	9.5	9.5	10.4	10.6	10.3	9.1	8.4	0.93	0.73-1.17	.517	0.99	0.81-1.22	.920	Reference
12-14 y	13.2	12.3	11.8	11.1	11.7	11.5	10.5	9.3	8.5	0.66	0.53-0.83	<.001	0.72	0.58-0.89	.002	<.001
15-17 y	11.3	10.5	10.0	9.1	9.1	8.7	7.7	6.6	6.2	0.53	0.41-0.70	<.001	0.58	0.45-0.75	<.001	<.001
Pervasive developmental disorders																
<12 y	4.8	5.3	3.7	5.3	4.8	4.5	3.8	4.3	4.6	0.87	0.35-2.18	.769	0.89	0.36-2.24	.810	Reference
12-14 y	3.3	3.6	3.0	4.0	3.8	3.7	2.9	3.3	3.2	0.92	0.58-1.46	.720	1.00	0.62-1.59	.987	.729
15-17 y	2.4	2.6	2.4	2.9	2.9	3.1	2.6	2.9	2.9	1.17	0.76-1.78	.481	1.29	0.84-1.99	.239	.244

Note: Analyses are based on the Mental Health Client-Level Data, 2013-2021. ADHD = attention-deficit/hyperactivity disorder; AOR = adjusted odds ratio; OR = odds ratio.

^aTrends were examined using logistic regression models adjusting for sex and race/ethnicity.

TABLE 4 Trends in Prevalence of Different Mental Disorders in Children and Adolescents Receiving Treatment in Community-Based Programs Between 2013 and 2021 According to Sex

	Percent of children and adolescent with each disorder receiving care in 2013-2021										Trend across 2013-2021 period						
	2013	2014	2015	2016	2017	2018	2019	2020	2021	Unadjusted			Adjusted ^a				
										OR	95% CI	p	AOR	95% CI	p	Interaction p	
Anxiety disorders																	
Female	11.7	12.5	14.4	15.3	18.1	20.0	21.3	21.3	24.2	2.41	2.04-2.84	<.001	2.41	2.05-2.83	<.001	Reference	
Male	8.2	8.6	10.0	10.2	11.8	12.8	13.7	13.4	15.0	1.97	1.65-2.35	<.001	1.96	1.65-2.32	<.001	<.001	
Depressive disorders																	
Female	18.8	19.7	21.5	20.5	20.7	21.9	23.2	22.3	24.0	1.30	1.13-1.51	<.001	1.24	1.05-1.46	.011	Reference	
Male	9.6	9.5	10.3	9.3	9.2	9.8	11.0	10.3	10.8	1.15	0.99-1.35	.069	1.14	0.98-1.34	.094	.002	
Bipolar disorders																	
Female	10.6	10.3	9.1	5.5	2.8	2.3	1.9	1.6	1.5	0.08	0.06-0.11	<.001	0.08	0.06-0.10	<.001	Reference	
Male	9.4	8.9	8.3	4.9	2.4	1.8	1.4	1.2	1.1	0.07	0.05-0.09	<.001	0.07	0.05-0.09	<.001	.010	
Trauma- and stressor-related disorders																	
Female	28.5	27.8	27.9	27.4	28.3	29.9	31.1	29.8	31.0	1.18	0.97-1.42	.091	1.20	0.99-1.45	.064	Reference	
Male	18.8	18.7	19.8	19.6	20.7	22.3	23.9	22.9	24.3	1.44	1.21-1.71	<.001	1.43	1.20-1.71	<.001	<.001	
ADHD																	
Female	19.9	19.0	17.6	18.0	18.5	18.7	17.0	15.9	16.0	0.79	0.67-0.92	.003	0.86	0.75-0.99	.040	Reference	
Male	36.8	35.9	34.6	34.9	36.5	37.2	34.6	32.7	33.6	0.88	0.73-1.07	.209	0.93	0.78-1.10	.392	<.001	
Conduct disorder																	
Female	7.0	6.7	6.3	5.3	4.5	3.9	3.5	3.0	2.7	0.35	0.27-0.45	<.001	0.37	0.28-0.48	<.001	Reference	
Male	11.5	11.3	11.0	9.5	8.4	7.6	7.1	6.3	5.8	0.45	0.34-0.58	<.001	0.45	0.34-0.60	<.001	<.001	
Oppositional defiant disorder																	
Female	9.0	8.4	7.8	7.5	7.7	7.5	6.9	5.9	5.3	0.62	0.48-0.78	<.001	0.66	0.53-0.82	<.001	Reference	
Male	12.6	12.0	12.0	11.6	12.4	12.5	11.9	10.6	10.0	0.84	0.67-1.05	.127	0.88	0.71-1.08	.208	<.001	
Pervasive developmental disorders																	
Female	1.7	1.9	1.4	1.9	1.8	1.7	1.5	1.7	1.7	0.92	0.44-1.94	.831	0.96	0.46-2.03	.924	Reference	
Male	5.2	5.9	4.6	6.1	5.8	5.6	4.7	5.3	5.6	0.98	0.48-1.99	.949	0.99	0.49-2.00	.968	.263	

Note: Analyses are based on Mental Health Client Level Data, 2013-2021. ADHD = attention-deficit/hyperactivity disorder; AOR = adjusted odds ratio; OR = odds ratio.

^aTrends were examined using logistic regression models adjusting for race/ethnicity and age group.

TABLE 5 Trends in Prevalence of Different Mental Disorders in Children and Adolescents Receiving Treatment in Community-Based Programs Between 2013 and 2021 According to Race/Ethnicity

	Percent of children and adolescents with each disorder receiving care in 2013-2021									Trend across 2013-2021 period						
	2013	2014	2015	2016	2017	2018	2019	2020	2021	Unadjusted			Adjusted ^a			
										OR	95% CI	p	AOR	95% CI	p	Interaction p
Anxiety disorders																
Hispanic	11.0	11.4	11.9	12.4	13.4	13.7	15.0	14.9	17.8	1.69	1.31-2.18	<.001	1.64	1.28-2.10	<.001	.001
Non-Hispanic Black	5.1	5.4	6.4	6.4	7.7	8.7	9.5	9.7	10.8	2.31	1.93-2.77	<.001	2.28	1.91-2.74	<.001	.419
Non-Hispanic White	11.1	11.8	14.2	14.7	17.2	18.8	20.5	20.5	23.5	2.45	2.07-2.90	<.001	2.41	2.03-2.87	<.001	Reference
Depressive disorders																
Hispanic	15.2	15.4	15.6	14.5	14.6	15.0	16.7	15.2	16.8	1.10	0.82-1.49	.518	1.01	0.74-1.38	.966	.047
Non-Hispanic Black	10.0	10.4	11.2	10.3	10.3	11.0	12.6	11.9	13.0	1.31	1.11-1.54	.001	1.29	1.09-1.53	.003	.537
Non-Hispanic White	13.0	13.6	15.4	14.3	14.8	15.6	16.9	16.3	18.1	1.41	1.24-1.59	<.001	1.36	1.18-1.56	<.001	Reference
Bipolar disorders																
Hispanic	10.7	9.8	7.7	3.9	1.9	1.6	1.3	1.2	1.2	0.06	0.04-0.07	<.001	0.05	0.04-0.07	<.001	.003
Non-Hispanic Black	9.5	9.3	8.9	5.1	2.6	2.1	1.7	1.6	1.4	0.09	0.06-0.14	<.001	0.09	0.06-0.13	<.001	.653
Non-Hispanic White	10.8	10.5	9.8	5.8	3.1	2.4	1.9	1.7	1.5	0.09	0.06-0.12	<.001	0.08	0.06-0.11	<.001	Reference
Trauma- and stressor-related disorders																
Hispanic	25.2	25.2	24.6	25.0	23.9	23.0	26.2	25.7	27.3	1.10	0.84-1.44	.483	1.09	0.82-1.44	.558	.158
Non-Hispanic Black	17.9	17.7	18.4	18.2	20.4	22.4	24.1	24.2	24.8	1.66	1.38-2.00	<.001	1.65	1.37-1.98	<.001	.002
Non-Hispanic White	23.8	23.7	25.2	24.5	25.4	27.1	29.0	27.3	28.5	1.32	1.12-1.56	.001	1.32	1.12-1.56	.001	Reference
ADHD																
Hispanic	29.9	27.4	23.2	24.7	25.6	24.7	21.1	21.5	20.8	0.66	0.51-0.84	.001	0.68	0.52-0.90	.006	.007
Non-Hispanic Black	36.9	34.6	35.0	35.8	37.1	37.5	35.2	35.1	35.1	0.98	0.81-1.18	.853	1.00	0.82-1.21	.994	.492
Non-Hispanic White	30.3	29.6	29.0	28.6	29.4	30.5	28.6	27.3	28.3	0.92	0.77-1.09	.318	0.95	0.79-1.13	.548	Reference
Conduct disorder																
Hispanic	8.7	7.6	6.6	5.9	5.1	4.5	4.2	3.4	3.0	0.33	0.23-0.49	<.001	0.35	0.24-0.51	<.001	.908
Non-Hispanic Black	12.9	12.5	12.2	10.2	9.6	8.8	8.0	7.8	6.8	0.48	0.39-0.60	<.001	0.49	0.40-0.60	<.001	.080
Non-Hispanic White	8.2	8.0	7.7	6.5	5.7	5.2	5.0	4.7	4.4	0.48	0.37-0.62	<.001	0.49	0.38-0.64	<.001	Reference
Oppositional defiant disorder																
Hispanic	9.9	8.7	7.3	7.8	8.0	7.6	6.9	5.6	5.0	0.55	0.39-0.78	.001	0.56	0.39-0.80	.001	.012
Non-Hispanic Black	15.3	14.3	14.6	13.9	14.7	14.6	13.7	13.8	12.4	0.86	0.62-1.19	.366	0.87	0.62-1.20	.391	.975
Non-Hispanic White	10.4	10.0	10.1	9.6	10.1	10.2	9.8	8.9	8.6	0.86	0.70-1.06	.152	0.88	0.72-1.08	.209	Reference
Pervasive developmental disorders																
Hispanic	3.0	3.1	2.8	3.0	3.0	3.3	2.5	4.0	4.2	1.42	1.00-2.03	.051	1.53	1.07-2.19	.019	.011
Non-Hispanic Black	1.9	2.2	1.7	2.5	2.5	2.7	2.4	2.6	2.9	1.46	0.54-3.96	.451	1.50	0.55-4.11	.425	<.001
Non-Hispanic White	5.0	5.7	4.2	5.7	5.2	4.9	4.1	4.0	4.0	0.73	0.35-1.55	.417	0.76	0.36-1.62	.483	Reference

Note: Analyses are based on Mental Health Client Level Data, 2013-2021. ADHD = attention-deficit/hyperactivity disorder; AOR = adjusted odds ratio; OR = odds ratio.

^aTrends were examined using logistic regression models adjusting for sex and age group.

treatment for these disorders. This trend was more pronounced in non-Hispanic Black children and adolescents than non-Hispanic White children and adolescents. In turn, the decrease in the diagnoses of bipolar disorders and oppositional defiant disorder was more pronounced in Hispanic children and adolescents than non-Hispanic White children and adolescents. Furthermore, Hispanic children and adolescents were the only racial/ethnic group that experienced a declining trend in ADHD and oppositional defiant disorder diagnoses and an increasing trend in pervasive developmental disorders.

Time trends according to the state median income level are presented in Table S3, available online. Trends appeared to be mostly consistent across states categorized according to median income level with the exception of the increasing trend in depressive disorders that was limited to states with low- and medium-income level. There was no significant increase in depressive disorders in high-income states. An increasing trend in pervasive developmental disorders limited to low-income states was also noted in these stratified analyses.

Sensitivity Analysis

Results of the analyses of trends across settings and in community-based programs after excluding states with missing data for the majority of years between 2013 and 2021 (ie, Alaska, Georgia, Kansas, Maine, New Hampshire, and West Virginia) or with partial data (New York and Tennessee) were quite consistent with the main analyses (Table S4, available online). Analyses based on the first recorded diagnosis were also mostly consistent with the main analyses (Table S5, available online). However, the increase in the proportion of depressive disorders did not reach a statistically significant level in adjusted analyses based on the first recorded diagnosis. Furthermore, change in trauma- and stressor-related disorders reached only a statistical trend level (AOR 1.18, 95% CI 0.99-1.41, $p = .061$ in all settings; AOR 1.18, 95% CI 0.99-1.40, $p = .060$ in community-based programs). However, the decrease in the proportion of children and adolescents with a diagnosis of ADHD became significant in models based on the first recorded diagnosis (AOR 0.85, 95% CI 0.74-0.97, $p = .020$ in all settings; AOR 0.85, 95% CI 0.74-0.97, $p = .016$ in community-based programs).

None of the quadratic terms for year reached a statistically significant level (data not shown), suggesting that the linear models adequately represented the trends. However, analyses to examine change in level and slope of the trends after the COVID-19 pandemic (2013-2019 vs

2020-2021) identified a statistically significant change in slope for trend in bipolar disorders ($\chi^2_1 = 13.20$, $p < .001$ across all settings; $\chi^2_1 = 11.95$, $p < .001$ in community-based programs). After a sharp decline in the 2013-2019 period, the decreasing trend became less steep in the 2020-2021 period.

DISCUSSION

This study examined trends in psychiatric diagnoses of children and adolescents receiving treatment in the state mental health system between 2013 and 2021. Significant increases occurred in the treatment of anxiety, depressive, and trauma- and stressor-related disorders in conjunction with reductions in bipolar, conduct, and oppositional defiant disorders. These findings extend to the public mental health system similar trends previously reported from other treatment settings and populations. For example, an increase occurred among child and adolescent anxiety, depression, and trauma- and stressor-related inpatient discharges within US community hospitals between 2009 and 2019.¹⁸ A claims analysis of outpatient commercial insurance claims further reported an increase in children and adolescents diagnosed with depression and anxiety disorders between 2018 and 2021.²¹

These trends in clinical diagnoses are also consistent with some, but not all,²¹ epidemiological trends in depression and anxiety disorders and externalizing problems as well as substance use disorders in general population surveys of adolescents.^{1-6,12-15} According to the National Survey on Drug Use and Health, for example, the percentage of adolescents ages 12-17 with past-year major depressive episodes increased from 8.1% (2009) to 15.8% (2019),³³ whereas the percentage with substance use disorders declined from 7.0% (2011) to 4.7% (2019).³⁴ It is therefore reasonable to assume that trends in clinical diagnoses at least partially reflect trends in prevalence of these conditions in the population and are not entirely a result of changes in treatment-seeking behavior, clinical diagnostic practices, or changes in diagnostic systems.

Some of the age, sex, and racial/ethnic variations in trends are also consistent with epidemiological trends. For instance, consistent with past reports, trends in depressive disorders were more pronounced in older adolescents, girls, and non-Hispanic White adolescents.¹ As another example, consistent with our findings, a study of change in prevalence of autism spectrum disorder diagnoses between 2014 and 2019 using the National Health Interview Survey found

statistically trend-level increases in the treated prevalence of this disorder among non-Hispanic Black and Hispanic children in later years, but no appreciable change in non-Hispanic White children.³⁵

Some of the observed trends may also be influenced by changes in the definition of disorders, detection of disorders, and patterns of treatment seeking. For instance, it is possible that the 2009 US Preventive Services Task Force (USPSTF) recommendation to screen adolescents for major depressive disorder in routine care may have contributed to the growing trends in depressive disorders observed in this study.³⁶ More recently, in 2022, the USPSTF recommended screening for anxiety in children and adolescents ages 8 to 18 years.³⁷

Although little is known about the quality of care of children and adolescents with depression or anxiety within the state mental health system, the quality of Medicaid-financed care appears to be uneven. In an analysis of Medicaid-covered youth ages 5 to 17 years with new episodes of major depressive disorder, 38.3% received minimally adequate psychotherapy, and 19.2% received minimally adequate pharmacotherapy.³⁸ The increasing percentage of children and adolescents in the state mental health system diagnosed with depressive or anxiety disorders should bring renewed attention to ensuring that these young patients receive high-quality evidence-based treatment.

The most pronounced trend in our analyses was the marked decline in children and adolescents with a diagnosis of bipolar disorder receiving treatment. This trend was also the most consistent across different demographic groups and treatment settings. In contrast to this finding, another study based on MarketScan commercial claims data revealed no change in the treated prevalence of pediatric bipolar disorder between 2012 and 2018.²² However, consistent with our finding, a study of acute inpatient admissions in the United States also recorded a decreasing trend in admissions with bipolar diagnoses between 2009 and 2019.¹⁹ The prevalence of outpatient treatment for bipolar disorder in children and adolescents increased 40-fold between 1994-1995 and 2002-2003.³⁹ This trend raised concerns about the accuracy of clinical diagnosis of pediatric bipolar disorder³⁹ and led to attempts to impose stricter delineation of bipolar disorder in children and adolescents, including through introduction of disruptive mood dysregulation disorder in *DSM-5* and removing “nonepisodic irritability” from the criteria for childhood mania.⁴⁰ Although the impact of these changes in diagnostic systems on diagnostic practices of providers is not clear, the nearly 8-fold

decrease in the proportion with bipolar disorder in the present study could reflect changes in diagnostic practices resulting from concerns about overdiagnosis of pediatric bipolar disorder.⁴¹

The increasing trend in trauma- and stressor-related disorders is somewhat puzzling, as the prevalence of several adverse childhood experiences appears to have decreased in the United States over the past 2 decades.⁴² It is plausible that heightened public attention and reduced stigma concerning childhood trauma may have motivated a greater number of children to report these experiences.⁴³ It is also conceivable that the growing popularity of trauma-focused care approaches in clinical settings⁴⁴ may have contributed to greater clinical awareness and recognition of trauma- and stressor-related disorders in these settings.

In interpreting the findings of this study, the limitations of these administrative state-level data should be considered. First, the study was based on clinician diagnoses rather than research diagnoses, and therefore the recorded diagnoses are open to practice style variations and are not subject to reliability assessment or expert validation. Second, information regarding the specific type of treatment (therapy, medications), length of treatment episodes, and outcomes of treatment was not available. Third, data on only a limited number of sociodemographic characteristics were available. Important information including residential ZIP code, household structure, household income, history of child neglect or abuse, and social determinants of health was not available. Fourth, there is growing concern about self-harm behavior and suicide deaths in youth.^{45,46} However, MH-CLD does not collect data on suicidal behavior. Fifth, broad age bands prevented analysis of diagnostic patterns among younger children, when several psychiatric disorders typically have their onset.⁴⁷ Lastly, there was a marked change in delivery of outpatient mental health services from in person to virtual following the COVID-19 pandemic.^{48,49} However, visit modality was not captured in MH-CLD.

In the context of these limitations, the findings from this study present a broad overview of trends in psychiatric diagnoses of children and adolescents receiving care in the public mental health care system. The findings highlight an increase in the treatment of children and adolescents diagnosed with anxiety, depressive, or trauma- and stressor-related disorders and a decrease in bipolar, conduct, and oppositional defiant disorders across demographic groups. The results do not permit distinguishing the contributions of changes in population prevalence from changes in treatment-seeking or diagnostic practices. The findings also

underscore the increasing importance of training the public child and adolescent mental healthcare workforce in evidence-supported assessment and management of increasingly prevalent anxiety, depressive, and trauma- and stressor-related disorders in these settings.

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